Curriculum Vitae

Gokarna Chand

Graduate student in Civil and Environmental Engineering in HKUST, with keen interest in the biology of microbial community in anaerobic digestion (AD) system, centered on improving stability and performance of AD processes through the integration of nanobubble technology.

Contact

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The Hong Kong University of Science and Technology
Clear Water Bay, Kowloon
Hong Kong

Gokarna Chand

Education

MPhil in Civil and Environmental Engineering (Currently)
(Expected Feb 2028)
The Hong Kong University of Science and Technology

B.E. in Chemical Engineering
(Oct 2024)
Tribhuvan University, Nepal
GPA: 77.96/100

Skills & Expertise

- Electro-chemical analysis (CV, GCD, ED, PD)
- Pyrolyzer, SEM, FT-IR
- DWSIM, ChemDraw
- AutoCAD, Data Analysis
- Cellulose, hemi-cellulose and lignin extraction
- Reactor setup and calibration

Professional Experience

Feb 2026–Current
Graduate Research Assistant
Department of Civil and Environmental Engineering, HKUST
Research focus: DIET in AD system.
PI: Prof. Samir Kumar Khanal

Apr 2024–Sep 2025
Undergraduate Research Assistant
Department of Applied Sciences and Chemical Engineering, Pulchowk Campus, TU, Nepal

- Role: Conducted electro-chemical analysis of biomass.
- PI: Dr. Tanka Mukhiya & Dr. Sahira Joshi

Apr 2023–Apr 2024
Undergraduate Research Assistant
Department of Applied Sciences and Chemical Engineering. Pulchowk Campus, TU, Nepal

- Role: Operated pyrolyzer and performed electro-chemical analysis of carbonized paper
- PI: Prof. Hem Raj Pant

Nov 2023–Dec 2023
Undergraduate Research Intern
Indian Institute of Technology (IIT), Roorkee, India

- Role: Extraction of cellulose, hemi-cellulose and lignin and FESEM analysis.
- PI: V.C. Srivastava

Research Areas

- Anaerobic Digestion (AD) system
- Nanobubble technology
- Pyrolysis
- Extraction of lignocellulosic biomass

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Awards and Honors

1. **PostGraduate Studentship**–The Hong Kong University of Science and Technology–Feb 2026–Current
2. **Best Poster Award**–International Conference on Advanced Functional Materials–Oct 2024

Conference Presentation (* denotes presenter)

1. **Chand, G.***, Kamal, A., & Paudel, Y. Pyrolytic Conversion of *Daphne bholua* For the Fabrication of Free-Standing Carbon Electrodes.Oct 27, 2024, International Conference on Advanced Functional Materials, Kathmandu, Nepal

Certifications

1. Responsible Conduct of Research (1-RCR), The Hong Kong University of Science and Technology, Apr 2026
2. Lab Safety Training on Chemical, Biological and Fire Safety, Feb 2026
3. Assistant Manager, International Conference on Advanced Functional Materials (ICAFM), Oct 2024

Relevant Courses

Graduate Courses
(MPhil in Civil and Environmental Engineering)

- Resource Recovery from Bioresources (CIVL4100P)
- Aquatic Chemistry (CIVL 5430)

Undergraduate Courses
(B.E. in Chemical Engineering)

- Transport Phenomena (EC756)
- Biochemical Engineering (EC752)
- Maintenance Engineering and Safety (EC707)
- Modelling and Simulation in Chemical Engineering (EC706)
- Chemical Engineering Design I/II (EC651/EC701)
- Mass Transfer I/II (EC602/EC654)
- Chemical Reaction Engineering I/II (EC606/EC655)
- Process Economics and Plant Design (EC708)
- Chemical Process Calculation (EC504)
- Computer Aided Drawing (ME454)
- Chemical Engineering Thermodynamics I/II (EC551/EC601)
- Mechanical Operation (EC604)
- Instrumentation and Process Control (EC652)
- Chemical Process Industries I/II (EC552/EC603)